

Short Answer

1. Briefly describe the difference between the prefix and postfix modes used by the increment and decrement operators.
2. Why should you indent the statements in the body of a loop?
3. Describe the difference between pretest loops and posttest loops.
4. Why are the statements in the body of a loop called conditionally executed statements?
5. Describe the difference between the `while` loop and the `do-while` loop.
6. Which loop should you use in situations where you want the loop to repeat until the boolean expression is `false`, and the loop should not execute if the test expression is `false` to begin with?
7. Which loop should you use in situations where you want the loop to repeat until the boolean expression is `false`, but the loop should execute at least once?
8. Which loop should you use when you know the number of required iterations?
9. Why is it critical that accumulator variables are properly initialized?
10. What is an infinite loop? Write the code for an infinite loop.
11. Describe a programming problem that would require the use of an accumulator.
12. What does it mean to let the user control a loop?
13. What is the advantage of using a sentinel?
14. Why must the value chosen for use as a sentinel be carefully selected?
15. Describe a programming problem requiring the use of nested loops.
16. How does a file buffer increase a program's performance?
17. Why should a program close a file when it's finished using it?
18. What is a file's read position? Where is the read position when a file is first opened for reading?
19. When writing data to a file, what is the difference between the `print` and the `println` methods?
20. What does the `Scanner` class's `hasNext` method return when the end of the file has been reached?
21. What is a potential error that can occur when a file is opened for reading?
22. What does it mean to append data to a file?
23. How do you open a file so that new data will be written to the end of the file's existing data?

Programming Challenges

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1. Sum of Numbers

Write a program that asks the user for a positive nonzero integer value. The program should use a loop to get the sum of all the integers from 1 up to the number entered. For example, if the user enters 50, the loop will find the sum of 1, 2, 3, 4, . . . 50.

2. Distance Traveled

The distance a vehicle travels can be calculated as follows:

$$\text{Distance} = \text{Speed} * \text{Time}$$

For example, if a train travels 40 miles-per-hour for three hours, the distance traveled is 120 miles. Write a program that asks for the speed of a vehicle (in miles-per-hour) and the number of hours it has traveled. It should use a loop to display the distance a vehicle has traveled for each hour of a time period specified by the user. For example, if a vehicle is traveling at 40 mph for a three-hour time period, it should display a report similar to the one that follows:

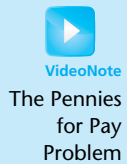
Hour	Distance Traveled
1	40
2	80
3	120

Input Validation: Do not accept a negative number for speed and do not accept any value less than 1 for time traveled.

3. Distance File

Modify the program you wrote for Programming Challenge 2 (Distance Traveled) so it writes the report to a file instead of the screen. Open the file in Notepad or another text editor to confirm the output.

4. Pennies for Pay



Write a program that calculates the amount a person would earn over a period of time if his or her salary is one penny the first day, two pennies the second day, and continues to double each day. The program should display a table showing the salary for each day, and then show the total pay at the end of the period. The output should be displayed in a dollar amount, not the number of pennies.

Input Validation: Do not accept a number less than 1 for the number of days worked.

5. Letter Counter

Write a program that asks the user to enter a string, and then asks the user to enter a character. The program should count and display the number of times that the specified character appears in the string.

6. File Letter Counter

Write a program that asks the user to enter the name of a file, and then asks the user to enter a character. The program should count and display the number of times that the specified character appears in the file. Use Notepad or another text editor to create a simple file that can be used to test the program.

7. Hotel Occupancy

A hotel's occupancy rate is calculated as follows:

$$\text{Occupancy rate} = \text{Number of rooms occupied} \div \text{Total number of rooms}$$

Write a program that calculates the occupancy rate for each floor of a hotel. The program should start by asking for the number of floors in the hotel. A loop should then iterate once for each floor. During each iteration, the loop should ask the user for the number of rooms on the floor and the number of them that are occupied. After all the iterations, the program should display the number of rooms the hotel has, the number of them that are occupied, the number that are vacant, and the occupancy rate for the hotel.

Input Validation: Do not accept a value less than 1 for the number of floors. Do not accept a number less than 10 for the number of rooms on a floor.

8. Average Rainfall

Write a program that uses nested loops to collect data and calculate the average rainfall over a period of years. First the program should ask for the number of years. The outer loop will iterate once for each year. The inner loop will iterate 12 times, once for each month. Each iteration of the inner loop will ask the user for the inches of rainfall for that month. After all iterations, the program should display the number of months, the total inches of rainfall, and the average rainfall per month for the entire period.

Input Validation: Do not accept a number less than 1 for the number of years. Do not accept negative numbers for the monthly rainfall.

9. Population

Write a program that will predict the size of a population of organisms. The program should ask for the starting number of organisms, their average daily population increase (as a percentage), and the number of days they will multiply. For example, a population might begin with two organisms, have an average daily increase of 50 percent, and will be allowed to multiply for seven days. The program should use a loop to display the size of the population for each day.

Input Validation: Do not accept a number less than 2 for the starting size of the population. Do not accept a negative number for average daily population increase. Do not accept a number less than 1 for the number of days they will multiply.

10. Largest and Smallest

Write a program with a loop that lets the user enter a series of integers. The user should enter -99 to signal the end of the series. After all the numbers have been entered, the program should display the largest and smallest numbers entered.

11. Celsius to Fahrenheit Table

Write a program that displays a table of the Celsius temperatures 0 through 20 and their Fahrenheit equivalents. The formula for converting a temperature from Celsius to Fahrenheit is

$$F = \frac{9}{5}C + 32$$

where F is the Fahrenheit temperature and C is the Celsius temperature. Your program must use a loop to display the table.

12. Bar Chart

Write a program that asks the user to enter today's sales for five stores. The program should display a bar chart comparing each store's sales. Create each bar in the bar chart by displaying a row of asterisks. Each asterisk should represent \$100 of sales. Here is an example of the program's output:

```
Enter today's sales for store 1: 1000 [Enter]
Enter today's sales for store 2: 1200 [Enter]
Enter today's sales for store 3: 1800 [Enter]
Enter today's sales for store 4: 800 [Enter]
Enter today's sales for store 5: 1900 [Enter]
```

```
SALES BAR CHART
Store 1: *****
Store 2: *****
Store 3: *****
Store 4: *****
Store 5: *****
```

13. File Head Display

Write a program that asks the user for the name of a file. The program should display only the first five lines of the file's contents. If the file contains fewer than five lines, it should display the file's entire contents.

14. Line Numbers

Write a program that asks the user for the name of a file. The program should display the contents of the file with each line preceded with a line number followed by a colon. The line numbering should start at 1.

15. Uppercase File Converter

Write a program that asks the user for the names of two files. The first file should be opened for reading and the second file should be opened for writing. The program should read the contents of the first file, change all characters to uppercase, and store the results in the second file. The second file will be a copy of the first file, except that all the characters will be uppercase. Use Notepad or another text editor to create a simple file that can be used to test the program.

16. Budget Analysis

Write a program that asks the user to enter the amount that he or she has budgeted for a month. A loop should then prompt the user to enter each of his or her expenses for the month, and keep a running total. When the loop finishes, the program should display the amount that the user is over or under budget.

17. Random Number Guessing Game

Write a program that generates a random number and asks the user to guess what the number is. If the user's guess is higher than the random number, the program should display "Too high, try again." If the user's guess is lower than the random number, the program should display "Too low, try again." The program should use a loop that repeats until the user correctly guesses the random number.

18. Random Number Guessing Game Enhancement

Enhance the program that you wrote for Programming Challenge 17 so it keeps a count of the number of guesses that the user makes. When the user correctly guesses the random number, the program should display the number of guesses.

19. ESP Game

Write a program that tests your ESP (extrasensory perception). The program should randomly select the name of a color from the following list of words:

Red, Green, Blue, Orange, Yellow

To select a word, the program can generate a random number. For example, if the number is 0, the selected word is *Red*, if the number is 1, the selected word is *Green*, and so forth.

Next, the program should ask the user to enter the color that the computer has selected. After the user has entered his or her guess, the program should display the name of the randomly selected color. The program should repeat this 10 times and then display the number of times the user correctly guessed the selected color.

20. Square Display

Write a program that asks the user for a positive integer no greater than 15. The program should then display a square on the screen using the character 'X'. The number entered by the user will be the length of each side of the square. For example, if the user enters 5, the program should display the following:

```
XXXXX
XXXXX
XXXXX
XXXXX
XXXXX
```

If the user enters 8, the program should display the following:

```
XXXXXXXX
XXXXXXXX
XXXXXXXX
XXXXXXXX
XXXXXXXX
XXXXXXXX
XXXXXXXX
XXXXXXXX
```

21. Dice Game

Write a program that plays a simple dice game between the computer and the user. When the program runs, a loop should repeat 10 times. Each iteration of the loop should do the following:

- Generate a random integer in the range of 1 through 6. This is the value of the computer's die.
- Generate another random integer in the range of 1 through 6. This is the value of the user's die.
- The die with the highest value wins. (In case of a tie, there is no winner for that particular roll of the dice.)

As the loop iterates, the program should keep count of the number of times the computer wins, and the number of times that the user wins. After the loop performs all of its iterations, the program should display who was the grand winner, the computer or the user.

22. Slot Machine Simulation

A slot machine is a gambling device that the user inserts money into and then pulls a lever (or presses a button). The slot machine then displays a set of random images. If two or more of the images match, the user wins an amount of money that the slot machine dispenses back to the user.

Create a program that simulates a slot machine. When the program runs, it should do the following:

- Asks the user to enter the amount of money he or she wants to enter into the slot machine.
- Instead of displaying images, the program will randomly select a word from the following list:

Cherries, Oranges, Plums, Bells, Melons, Bars

To select a word, the program can generate a random number in the range of 0 through 5. If the number is 0, the selected word is *Cherries*; if the number is 1, the selected word is *Oranges*; and so forth. The program should randomly select a word from this list three times and display all three of the words.

- If none of the randomly selected words match, the program will inform the user that he or she has won \$0. If two of the words match, the program will inform the user that he or she has won two times the amount entered. If three of the words match, the program will inform the user that he or she has won three times the amount entered.
- The program will ask whether the user wants to play again. If so, these steps are repeated. If not, the program displays the total amount of money entered into the slot machine and the total amount won.

Decision Structures